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22CS701- CRYPTOGRAPHY AND CYBER SECURITY

(Lab Integrated)

Unit-III

Department: Computer Science and Engineering

Batch/Year : 2023-2027/IV

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COLLEGE VISION & MISSION

VISION

To be a globally recognized institution of technical excellence that nurtures knowledge, fosters innovation, and promotes research aligned with the Sustainable Development Goals, transforming students into ethical engineers and visionary leaders for societal development.

MISSION

- To provide quality technical education that builds strong fundamentals, practical skills, and lifelong learning abilities in students.
- To promote innovation, entrepreneurship, and research through a creative academic environment and industry collaboration.
- To develop ethically responsible engineers who contribute to sustainable development and societal well-being. To encourage interdisciplinary learning and problem-solving that addresses real-world challenges in line with the Sustainable Development Goals.
- To nurture leadership, teamwork, and professional excellence so that students emerge as competent engineers and visionary leaders.

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DEPARTMENT VISION AND MISSION

VISION

To produce intellectual and globally competent software professionals through excellence in education, innovation, and research, providing sustainable solution for societal development.

MISSION

- To impart modern teaching methodologies to provide quality technical education in Computer Science and Engineering.
- To establish a conducive learning environment to practice emerging software tools and programming languages.
- To identify industry requirements and develop employable software engineers.
- To engage students and faculty in research and innovation to address the evolving needs of the software industry and societal development.

22CS701-CRYPTOGRAPHY AND CYBER SECURITY
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COURSE OBJECTIVES

The Course will enable learners to:

- Understand the fundamentals of network security and security architecture.
- Learn the different symmetric key cryptographic algorithms.
- Study the various asymmetric key cryptographic algorithms and techniques.
- Know the importance of message authentication and integrity.
- Learn the various cyber-crimes and cyber security

PREREQUISITE

- 22MA201 Transforms and Numerical Methods
- 22CS501 Computer Networks

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22CS701 CRYPTOGRAPHY AND CYBER SECURITY (Lab Integrated)

SYLLABUS 3 0 2 4

UNIT I INTRODUCTION TO SECURITY 9+6

Computer Security Concepts – The OSI Security Architecture – Security Attacks – Security Services and Mechanisms – A Model for Network Security – Classical encryption techniques: Substitution techniques, Transposition techniques, Steganography –Foundations of modern cryptography: Perfect security – Information Theory – Product Cryptosystem – Cryptanalysis.

List of Exercise/Experiments:

1. Perform encryption, decryption using the following substitution techniques (i) Ceaser cipher, (ii) playfair cipher iii) Hill Cipher iv) Vigenere cipher
2. Perform encryption and decryption using following transposition techniques i) Rail fence ii) row & Column Transformation

UNIT II SYMMETRIC CIPHERS 9+6

Number theory – Algebraic Structures – Modular Arithmetic - Euclid‘s algorithm – Congruence and matrices – Group, Rings, Fields, Finite Fields SYMMETRIC KEY CIPHERS: SDES – Block Ciphers – DES, Strength of DES – Differential and linear cryptanalysis – Block cipher design principles – Block cipher mode of operation – Evaluation criteria for AES – Pseudorandom Number Generators – RC4 – Key distribution.

List of Exercise/Experiments:

1. Apply DES algorithm for practical applications.
2. Apply AES algorithm for practical applications.

UNIT III ASYMMETRIC CRYPTOGRAPHY 9+6

MATHEMATICS OF ASYMMETRIC KEY CRYPTOGRAPHY: Primes – Primality Testing –Factorization – Euler’s totient function, Fermat’s and Euler’s Theorem – Chinese Remainder Theorem – Exponentiation and logarithm ASYMMETRIC KEY CIPHERS: RSA cryptosystem – Key distribution – Key management – Diffie Hellman key exchange -– Elliptic curve arithmetic – Elliptic curve cryptography.

List of Exercise/Experiments:

1. Implement RSA Algorithm using HTML and JavaScript.
2. Implement the Diffie-Hellman Key Exchange algorithm for a given problem.
3. Calculate the message digest of a text using the SHA-1 algorithm.

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SYLLABUS 3 0 2 4

UNIT IV INTEGRITY AND AUTHENTICATION ALGORITHMS 9+6

Authentication requirement – Authentication function – MAC – Hash function – Security of hash function: HMAC, CMAC – SHA – Digital signature and authentication protocols – DSS – Schnorr Digital Signature Scheme – ElGamal cryptosystem – Entity Authentication: Biometrics, Passwords, Challenge Response protocols – Authentication applications – Kerberos MUTUAL TRUST: Key management and distribution – Symmetric key distribution using symmetric and asymmetric encryption – Distribution of public keys – X.509 Certificates

List of Exercise/Experiments:

1. Implement the SIGNATURE SCHEME - Digital Signature Standard.
2. Demonstrate intrusion detection system (ids) using any tool eg. Snort or any other s/w

UNIT V CYBER CRIMES AND CYBER SECURITY 9+6

Cyber Crime and Information Security – classifications of Cyber Crimes – Tools and Methods – Password Cracking, Keyloggers, Spywares, SQL Injection – Network Access Control – Cloud Security – Web Security – Wireless Security

List of Exercise/Experiments:

1. Automated Attack and Penetration Tools

a. Exploring N-Stalker, a Vulnerability Assessment Tool

2. Defeating Malware

i) Building Trojans ii) Rootkit Hunter

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COURSE OUTCOMES

CO1	Explore cryptographical concepts
CO2	Implement various cryptographic algorithms
CO3	Evaluate and apply network security protocols, to secure communications over networks
CO4	Identify common security threats and vulnerabilities and assess their impact on network security
CO5	Implement access control mechanisms and authentication techniques to protect information systems
CO6	Develop and propose security policies and best practices for securing networks and information systems

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CO-PO MAPPING

COs	PO's/PSO's														
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	3	3	2	2	-	-	2	2	1	-	2	2	2	1
CO2	3	3	3	2	2	-	-	2	2	1	-	2	2	2	1
CO3	3	3	3	2	2	-	-	2	2	1	-	2	2	2	1
CO4	3	3	3	2	2	-	-	2	2	1	-	2	2	2	1
CO5	3	3	3	2	2	-	-	2	2	1	-	2	2	2	1
CO6	3	3	3	2	2	-	-	2	2	1	-	2	2	2	1

1 – Low, 2 – Medium, 3 – Strong

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LECTURE PLAN

S No	Topics	No. of periods	Pertaining CO	Taxonomy level	Mode of delivery
1	Course objective, course outcome delivery & Course introduction	1	CO3	K1	ICT Tools
2	Primes , Primality Testing, Factorization	1	CO3	K3	ICT Tools
3	Euler's totient function, Fermat's and Euler's Theorem	1	CO3	K3	ICT Tools
4	Chinese Remainder Theorem – Exponentiation and logarithm	1	CO3	K3	ICT Tools
5	RSA cryptosystem	1	CO3	K3	ICT Tools
6	Key distribution, Key management	1	CO3	K2	ICT Tools
7	Diffie Hellman key exchange	1	CO3	K3	ICT Tools
8	Elliptic curve arithmetic	1	CO3	K3	ICT Tools
9	Elliptic curve cryptography	2	CO3	K3	ICT Tools

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PEDAGOGY

Topic	Suggested Pedagogy	Classroom Activity
Primes	Concept attainment + guided discovery	Prime Detective: groups classify given integers as prime/composite and justify the decision using suitable tests.
Primality Testing	Problem-based learning + peer instruction	Primality Test Challenge: students apply trial division and selected probabilistic ideas to determine whether given numbers are prime.
Factorization	Worked examples + collaborative problem solving	Factor Hunt: teams factor selected integers and compare the efficiency of different approaches.
Euler's Totient Function	Step-by-step modelling + practice	Totient Card Game: students calculate $\varphi(n)$ for prime, composite and relatively-prime cases and explain patterns.
Fermat's and Euler's Theorem	Think–Pair–Share + numerical verification	Theorem Verification Lab: students solve modular exponentiation examples and verify theorem conditions.

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CASE STUDY

S.NO	TOPICS	CASE STUDY	Learning outcome
1	RSA Cryptosystem	Build Your Own RSA: students generate small keys, perform encryption/decryption, and trace every step of the algorithm.	Explain RSA key generation and perform encryption/decryption.
2	Key Management	Key Lifecycle Mapping: groups map generation, storage, distribution, rotation, backup, revocation and destruction of keys.	Recognize the importance of secure key lifecycle management.
3	Elliptic Curve Cryptography	ECC vs RSA Discussion: groups compare key size, computational concepts, security assumptions and suitable applications.	Relate elliptic-curve arithmetic to public-key cryptography and compare ECC with RSA.

ACTIVITY BASED LEARNING

1. Diffie Hellman Key Exchange

Activity Title: *"Secret Key Without Sharing the Secret"*

Activity Type: Role Play + Hands-on Calculation + Simulation

Materials Required

- Whiteboard / chart
- Calculators (or mobile)
- Worksheets with modular arithmetic steps

Activity Procedure

- Step 1: Setup (Instructor Demonstration) – 10 mins
- Step 2: Role Play (Student Interaction) – 20 mins
- Step 3: Attacker Perspective – 10 mins
- Step 4: Reflection & Discussion – 10 mins

2. RSA Cryptosystem

Activity Title: *"Encrypt, Send, Decrypt – The RSA Mission"*

Activity Type: Role Play + Guided Calculation + Problem Solving

Materials Required

Worksheets (with step-by-step RSA process)

Calculator / mobile

Whiteboard

Activity Procedure

Step 1: Concept Introduction – 10 mins

Step 2: Group Role Play – 25 mins

Step 3: RSA Key Generation (Bob's Task)

Step 4: Encryption (Alice's Task) – 10 mins

Step 5: Decryption (Bob's Task) – 10 mins

Step 6: Attacker Perspective – 10 mins

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ACTIVITY BASED LEARNING

Euler's Totient Function

Activity Title: *"Coprime Counting Challenge"*

Activity Type: Group Work + Exploration + Pattern Discovery

Activity Steps

Step 1: Manual Counting (15 mins)

Step 2: Pattern Discovery (10 mins)

Step 3: Formula Application (10 mins)

Step 4: Cryptography Link (10 mins)

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MATHEMATICS OF ASYMMETRIC KEY CRYPTOGRAPHY

Prime

An integer p is said to be a prime if p>=2 and its only divisors are the trivial divisors 1 and p.

An integer greater than 2 that is not prime is said to be composite.

Examples of Prime: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, ...

Examples of Composite: 4, 6, 8, 9, 10, 12, 14, 15,...

Fermat's Theorem

If p is prime and a is a positive integer not divisible by p, then

$$a^{p-1} \equiv 1 \pmod{p}$$

An alternative form of Fermat's theorem is also useful: If is prime and is a positive integer, then $a^p \equiv p \pmod{p}$

Example:

$8^{19-1} \equiv 1 \bmod 19$

$8^2 \equiv 7 \bmod 19$ $8^4 \equiv 11 \bmod 19$ $8^8 \equiv 7 \bmod 19$ $8^{16} \equiv 11 \bmod 19$

$8^{18} \equiv 77 \equiv 1 \bmod 19$

Proof:

Consider a set of positive integers less than p: S = {1, 2, … p-1}

Another set X = {a mod p, 2a mod p, … (p-1)a mod p} which is multiple of a mod p the set S.

None of the elements of X is equal to zero because p does not divide a. All elements of X are unique.

S ≡ X mod p (Since X is permutation of S)

Multiplying the numbers in both sets (S and X) and taking the result mod yields

$[1 \times 2 \times \ldots (p-1)] \bmod p = [a \times 2a \times \ldots (p-1)] \bmod p$

$[a \times 2a \times \ldots (p-1)] \equiv [1 \times 2 \times \ldots (p-1)] \bmod p$

$a^{p-1} (p-1)! \equiv (p-1) ! \bmod p$

$a^{p-1} \equiv 1 \pmod{p}$

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MATHEMATICS OF ASYMMETRIC KEY CRYPTOGRAPHY

❁ Euler's Totient Function

It is defined as the number of positive integers less than n and relatively prime to n. It is denoted by φ (n).

Reduced Residue System modulo n

The set of positive integers less than n and relatively prime to n. (or) The set of all residue classes relatively prime to n.

Example:

Reduced Residue Set: {1, 3, 7, 9} φ (10) = 4. So, Euler's totient function gives the cardinality of the reduced residue set.

- ❖ φ (mn) = φ(m) . φ(n) if gcd (m, n) =1
- ❖ φ (p) = p -1 p is prime
- ❖ φ (p^j) = p^j - p^{j-1}
- ❖ Φ (n) = n $\prod_{p|n}(1-\frac{1}{p})$

Examples:

φ(97) = 96

Φ(343) = φ(7^3) = 7^3 - 7^2 = 343 - 49 = 294

φ (343) = 343. ∏ (1-1/7) = 343.6/7 = 294

φ (72) = φ (8) φ(9) = 4.6 = 24

❁ Euler's Theorem

If a and n are relatively prime, then $a^{\varphi(n)} \equiv 1 \pmod n$.

Alternative form of this theorem is $a^{\varphi(n)+1} \equiv a \pmod n$.

Proof:

φ (n) is the number of positive integers less than n and relatively prime to n.

Consider the set of such integers, labeled as

R = {x_1, x_2, …, $x_{\varphi(n)}$ }

S = {(ax_1 mod n), (ax_2 mod n), …, ($ax_{\varphi(n)}$ mod n)}

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The set S is a permutation of R, by the following line of reasoning:

1. Because *a* is relatively prime to *n* and x_i is relatively prime to n, ax_i must also be relatively prime to n. Thus, all the members of S are integers that are less than n and that are relatively prime to n.
2. There are no duplicates in S. If $ax_i \bmod n = ax_j \bmod n$, then $x_i = x_j$.

Therefore,

$$\prod_{i=1}^{\varphi(n)} (ax_i \bmod n) = \prod_{i=1}^{\varphi(n)} x_i$$

$$\prod_{i=1}^{\varphi(n)} ax_i \equiv \prod_{i=1}^{\varphi(n)} x_i \bmod n$$

$$a^{\varphi(n)} \times \left[\prod_{i=1}^{\varphi(n)} x_i\right] \equiv \prod_{i=1}^{\varphi(n)} x_i \bmod n$$

$$a^{\varphi(n)} \equiv 1 \pmod n$$

Example:

$6^{\varphi(11)} \equiv 1 \pmod{11}$ $\varphi(11) = 10$

$6^2 \equiv 3 \pmod{11}$

$6^4 \equiv 9 \pmod{11}$

$6^8 \equiv 4 \pmod{11}$

$6^{10} \equiv 4.3 \equiv 12 \equiv 1 \pmod{11}$

❁ Primality Testing

The following are the some of the popular algorithms used for primality algorithms:

- ❖ Fermat Primality Test
- ❖ Solovay-Strassen Primality Test
- ❖ Miller-Rabin Primality Test

All these algorithms are probabilistic and not deterministic. It means that these algorithms determines whether a number is composite or probably prime.

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Deterministic Algorithm: AKS Primality Test

This algorithm determines whether a number is prime or not with 100% probability.

But this algorithm is not as efficient as Miller-Rabin algorithm.

- **Miller-Rabin Primality Test**

1. Find integers k, q, with k > 0, q odd, so that ($n - 1 = 2^k q$);
2. Select a random integer a, $1 < a < n-1$;
3. if $a^q \bmod n = 1$ then return ("inconclusive");
4. for j = 0 to k -1 do
5. if $a^{2jq} \bmod n = n - 1$ then return ("inconclusive")
6. return("composite");

Example: n=29

$n-1 = 28 = 2^2 \times 7$ where k=2 and q=7. Let a=2.

Test 1: To check if $a^q \bmod n = 1$

$2^7 = 128 \equiv 12 \not\equiv 1$ (So Test 1 fails)

Test 2: To check for j = 0 to k -1 do if $a^{2jq} \bmod n = n - 1$

j=0, $2^{1\times7} \bmod 29 = 2^7 \bmod 29 = 12 \neq 28$

J=1, $2^{2\times7} \bmod 29 = 2^{14} \equiv 144 \equiv 28 \bmod 29$ (Test 2 passes)

Therefore 29 is probably prime.

Probability of a composite number passing Miller-Rabin Test is $(¼)^t$, where t is number of trials or tests with t different values of a. So for t=10, the probability that a nonprime number will pass all ten tests is less than 10^{-6}.

Strong Pseudoprime

If n is composite and passes the Miller-Rabin test for the base a, then n is called a strong pseudoprime to the base.

Example: Show that 2047 is a strong pseudoprime to the base 2.

Let n = 2047 a=2

$n-1 = 2046 = 1023 \times 2^1$

k=1, q=1023

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Test 1: $a^q \bmod n = 1$

$2^{1023} \bmod 2047$

$2^{11} \equiv 2048 \equiv 1 \bmod 2047$

$2^{11} \equiv 2^{110} \equiv 2^{990} \equiv 2^{1001} \equiv 2^{1012} \equiv 2^{1023} \equiv 1 \bmod 2047$

$2047 = 23 \times 89$

Since 2047, which is composite number, passes Miller Rabin Test for a=2 is strong pseudoprime to the base 2.

❁ Chinese Remainder Theorem

The CRT says it is possible to reconstruct integers in a certain range from their residues modulo a set of pairwise relatively prime moduli.

Let $\mathbf{M = \prod_{i=1}^{k} m_i}$

where the mi are pairwise relatively prime; that is, $\gcd(m_i, m_j) = 1$ for $1 \le i, j \le k$, and $i \ne j$.

We can represent any integer a in Z_m by a k-tuple whose elements are in Z_m using the following correspondence:

$$\mathbf{A \leftrightarrow (a_1, a_2, \ldots, a_k)}$$

where $A \varepsilon Z_M$, $a_i \varepsilon Z_m$, and $a_i = A \bmod m_i$ for $1 \le i \le k$. The CRT makes two assertions.

1. The mapping of above Equation is a one-to-one correspondence (called a **bijection**) between Z_M and the Cartesian product $Z_{m1} \times Z_{m2} \times \ldots \times Z_{mk}$. That is, for every integer a such that $0 \le a \le M$, there is unique k-tuple $(a_1, a_2, \ldots a_k)$ with $0 \le a_i \le m_i$ that represents it, and for every such k-tuple $(a_1, a_2, \ldots a_k)$, there is a unique integer a in Z_M.
2. Operations performed on the elements of Z_M can be equivalently performed on the corresponding k-tuples by performing the operation independently in each coordinate position in the appropriate system.

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First Assertion:

$a_i = A \bmod m_i$ (each a_i is uniquely calculated)

Let $M_i = M/m_i$ for $1 \le i \le k$

$M = m_1 \times m_2 \times \ldots \times m_{i-1} \times m_{i+1} \times \ldots \times m_k$, so that $M_i=0 \pmod{m_j}$ for all $j \neq i$. Then let $\mathbf{c_i = M_i \times (M_i^{-1} \bmod m_i)}$ **for** $\mathbf{1 \le i \le k}$

By the definition of M_i, it is relatively prime to m_i and therefore has a unique multiplicative inverse mod m_i. So the above equation is well defined and produces a unique value. We can now compute

$A = (\sum_{i=1}^{k} a_i \, ci \,) \pmod{M}$

#Problem: Solve the following simultaneous linear congruences using CRT.

$X \equiv 1 \pmod 5$

$X \equiv 2 \pmod 7$

$X \equiv 3 \pmod 9$

Solution:

$a_1=1, a_2=2, a_3=3$ and $m_1 = 5, m_2 = 7, m_3 =9$

$M = 5 \times 7 \times 9 = 315$

$X = \Sigma \, a_i \, c_i \pmod M$

$c_i = M_i \times (M_i^{-1} \bmod m_i)$

$M_i = M/m_i$

$M_1 = 315/5 = 63$

$M_2 = 315/7 = 45$

$M_3 = 315/9 = 35$

$c1 = 63 \times (63^{-1} \bmod 5) = 63 \times (3^{-1} \bmod 5) = 63 \times 2 = 126$

$c2 = 45 \times (45^{-1} \bmod 7) = 45 \times (3^{-1} \bmod 7) = 45 \times 5 = 225$

$c3 = 35 \times (35^{-1} \bmod 9) = 35 \times (8^{-1} \bmod 9) = 35 \times 8 = 280$

$X = (1 \times 126 + 2 \times 225 + 3 \times 280) \bmod 315$

$X = (126+450+840) \bmod 315 = 1416 \bmod 315 = 156$

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Second Assertion:

It follows from rules for modular arithmetic.

If $A \leftrightarrow (a_1, a_2, ..., a_k)$ and $B \leftrightarrow (b_1, b_2, ..., b_k)$

then

$(A + B) \bmod M \leftrightarrow ((a_1 + b_1) \bmod m_1, ..., (a_k + b_k) \bmod m_k)$

$(A - B) \bmod M \leftrightarrow ((a_1 - b_1) \bmod m_1, ..., (a_k - b_k) \bmod m_k)$

$(A \times B) \bmod M \leftrightarrow ((a_1 \times b_1) \bmod m_1, ..., (a_k \times b_k) \bmod m_k)$

One of the useful features of the Chinese remainder theorem is that it provides a way to manipulate (potentially very large) numbers mod M in terms of tuples of smaller numbers.

Example:

To represent 959 mod 2491 as a pair of numbers mod 47 and 53.

Given: $m_1 = 47$ and $m_2 = 53$ $M = 2491$ $A = 959$

$M_1 = M/m_1 = 2491/47 = 53$
$M_{1_{-1}} = 53_{-1} \bmod 47 = 6_{-1} \bmod 47 = 8 \bmod 47$

$M_2 = M/m_2 = 2491/53 = 47$
$M_{2_{-1}} = 47_{-1} \equiv -9 \equiv 44 \bmod 53$

A1	A2	A3	B1	B2	B3	Q =(A3/B3)
1	0	53	0	1	47	1
0	1	47	1	-1	6	7
1	-1	6	-7	8	5	1
-7	8	5	8	-9	1	

959 is represented as (19, 5) as 959 mod 47 = 19 and 959 mod 53 = 5

To add 648 to 959,

$(648) \leftrightarrow (648 \bmod 47, 648 \bmod 53) = (37, 12)$

$(648+959) \bmod 2491 \leftrightarrow (37+19 \bmod 47, 12+5 \bmod 53) = (9, 17)$

To verify: $(9, 17) \leftrightarrow a_1M_1M_{1_{-1}} + a_2M_2M_{2_{-1}} \bmod M$

$= [(9)(53)(8) + (17)(47)(44)] \bmod 2491$

$= [3816 + 35156] \bmod 2491 = 1607$ $(648+959=1607) \bmod 2491$

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MATHEMATICS OF ASYMMETRIC KEY CRYPTOGRAPHY

Order of a (mod n)

If a and n are relatively prime, then there is atleast one integer m that satisfies

$a^m \equiv 1 \pmod n$ m is the smallest positive integer

The order of a modulo n is usually written as $m = ord_n(a)$

Example:

$7^1 \equiv 7 \bmod 19$ $7^2 \equiv 11 \bmod 19$ $7^3 \equiv 1 \bmod 19$

$m = ord_{19}(7) = 3$

The sequence is periodic $7^{3+j} \equiv 7^3 7^j \equiv 7^j \bmod 19$

$7^1 \equiv 7^4 \equiv 7^7 \equiv 7 \bmod 19$ $7^2 \equiv 7^5 \equiv 7^8 \equiv 11 \bmod 19$

Primitive Roots

Any base integer a which generates (via powers) the reduced residue class set modulo n, then such integer a is called as primitive root of modulus n.

If a is a primitive root of n, then its powers a, a^2, a^3, …, $a^{\varphi(n)}$ are distinct (mod n) and are all relatively prime to n. Moreover $a^{\varphi(n)} \equiv 1 \pmod n$

Example 1: 6 is primitive root of 11

$6^1 \equiv 6 \bmod 11$ $6^4 \equiv 9 \bmod 11$ $6^7 \equiv 8 \bmod 11$ $6^{10} \equiv 1 \bmod 11$

$6^2 \equiv 3 \bmod 11$ $6^5 \equiv 10 \bmod 11$ $6^8 \equiv 4 \bmod 11$

$6^3 \equiv 7 \bmod 11$ $6^6 \equiv 5 \bmod 11$ $6^9 \equiv 2 \bmod 11$

Reduced Residue Set of 11 = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

Example 2: 3 is primitive root of 10

$3^1 \equiv 3 \bmod 10$ $3^2 \equiv 9 \bmod 10$ $3^3 \equiv 7 \bmod 10$ $3^4 \equiv 1 \bmod 10$

Reduced Residue Set of 10 = {1, 3, 7, 9}

- The integers having primitive roots are 2, 4, p^n and $2p^n$, where p is any odd prime and n is a positive integer.
- The number of primitive roots modulo n, if there are any, is equal to ϕ ($\phi(n)$)

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Example: To check whether 6 is primitive root of 11

$\phi(n) = \phi(11) = 10 = 2.5$ (prime factors)

Test a=6:

$6^{10/5} = 6^2 \equiv 36 \equiv 3 \not\equiv 1 \bmod 11$

$6^{10/2} = 6^5 \equiv 3.3.6 \equiv 10 \not\equiv 1 \bmod 11$

6 is primitive root of 11

Discrete Logarithms (Index)

$b \equiv a^i \pmod p$ where $0 \le i \le (p-1)$

This exponent i is referred to as the discrete logarithm of the number b for the base a (mod p). This value is denoted as $\text{dlog}_{a,p}(b)$.

$\text{dlog}_{a,p}(1) = 0$ because $a^0 \bmod p = 1 \bmod p = 1$

$\text{dlog}_{a,p}(b) = 1$ because $a^1 \bmod p = b$

Methods to compute discrete logarithms:

- The Pohlig-Hellman Algorithm
- Baby Step, Giant Step
- The Index Calculus

Example:

$\text{dlog}_{2,11}(7) = ?$ Finding i such that $2^i \bmod 11 = 7$

Using Brute-force method:

$2 \bmod 11 = 2$ $2^2 \bmod 11 = 4$ $2^7 \bmod 11 = 7$

$2^3 \bmod 11 = 8$ $2^4 \bmod 11 = 5$

$2^5 \bmod 11 = 10$ $2^6 \bmod 11 = 9$

b	2	4	8	5	10	9	7
$\text{dlog}_{2,11}(b)$	1	2	3	4	5	6	7

Even though discrete logarithms is not a one-way trap-door function, it is a hard problem. That is, it is infeasible to compute discrete logarithms.

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ASYMMETRIC KEY CIPHERS

RSA

RSA was developed by Ron Rivest, Adi Shamir, and Len Adleman at MIT in 1977.

It is a public-key cryptographic algorithm.

The RSA scheme is a block cipher in which the plaintext and ciphertext are integers between 0 and n - 1 for some n.

A typical size for n is 1024 bits, or 309 decimal digits.

Applications of RSA

Key Exchange in TLS, WTLS etc.

Producing Digital Signatures in SSL Certificate, PGP, S/MIME etc.

Encrypting Symmetric Keys in PGP, S/MIME etc.

Algorithm Description

Let Plaintext block - M, Ciphertext block - C

$C = M^e \bmod n$

$M = C^d \bmod n = (M^e)^d \bmod n = M^{ed} \bmod n$

Both sender and receiver must know the value of n.

Sender knows the value of e and only receiver knows the value of d.

Public Key PU = {e, n}

Private Key PR = {d, n}

For this algorithm to be satisfactory for public-key encryption, the following requirements must be met.

1. It is possible to find values of e, d, n such that $M^{ed} \bmod n = M$ for all $M < n$.
2. It is relatively easy to calculate $M^e \bmod n$ and $C^d \bmod n$ for all values of $M < n$.

3.It is infeasible to determine d given e and n.

$\phi(n) = \phi(pq) = \phi(p)\,\phi(q) = (p-1)(q-1)$

$ed \bmod \phi(n) = 1$

$ed \equiv 1 \bmod \phi(n)$

$d \equiv e^{-1} \bmod \phi(n)$

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ASYMMETRIC KEY CIPHERS

RSA Algorithm

User B (Receiver):

1. Select p,q p and q both prime, $p \neq q$
2. Calculate $n = p * q$
3. Calculate $\phi(n) = (p - 1)(q - 1)$
4. Select integer e, $\gcd(\phi(n), e) = 1; 1 < e < \phi(n)$
5. Calculated d $d \equiv e^{-1} \bmod \phi(n)$

Public key PU = {e, n} Private key PR = {d, n}

Encryption by User A (Sender) with User B's Public Key:

Plaintext: $M < n$

Ciphertext: $C = M^e \bmod n$

Decryption by User B (Receiver) with User B's Private Key:

Ciphertext: C

Plaintext: $M = C^d \bmod n$

Decryption using CRT

$M = C^d \bmod n$

$V_p = C^d \bmod p$ $V_q = C^d \bmod q$

$V_p = C^{d \bmod (p-1)} \bmod p$ $V_q = C^{d \bmod (q-1)} \bmod q$

$X_p = q \times (q^{-1} \bmod p)$ $X_q = p \times (p^{-1} \bmod q)$

$M = (V_pX_p + V_qX_q) \bmod n$

Fast Modular Exponentiation: To compute $a^b \bmod n$ quickly

Example: $2^{560} \bmod 561 = 1$ (final value of f in the below table)

Initial value of f = 1, a=2, n = 561, b = 560 = $(1000110000)_2$

f = (f × f) mod n	1	4	16	256	460	412	166	67	1	**1**
B_i	1	0	0	0	1	1	0	0	0	0
f = (f × a) mod n	2	-	-	-	359	263	-	-	-	-

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ASYMMETRIC KEY CIPHERS

#Problem: Perform encryption and decryption for the parameters

p=3, q=11, e=7, M=5

Solution:

Encryption

$C = M^e \bmod n$

$n = p \times q = 3 \times 11 = 33$

$C = 5^7 \bmod 33$

$5^3 = 125 \equiv 26 \bmod 33$

$5^6 \equiv 676 \equiv 16 \bmod 33$

$5^7 \equiv 16.5 \equiv 80 \equiv 14 \bmod 33$

$C = 14$

Decryption

$M = C^d \bmod n$

$\phi(n) = (p - 1)(q - 1) = 2.10 = 20$

$d \equiv e^{-1} \bmod \phi(n) \Rightarrow d \equiv 7^{-1} \bmod 20 \Rightarrow d=3$

$M = 14^3 \bmod 33$

$14^2 = 196 \equiv 31 \bmod 33$ $\quad$ $14^3 \equiv 31.14 \equiv 434 \equiv 5 \bmod 33$

$M = 5$

Decryption Using CRT

C=14, p=3, q=11 n=33

$V_p = C^{d \bmod (p-1)} \bmod p$ $\quad$ $V_q = C^{d \bmod (q-1)} \bmod q$

$V_p = 14^{3 \bmod 2} \bmod 3$ $\quad$ $V_q = 14^{3 \bmod 10} \bmod 11$

$V_p = 14 \bmod 3 = 2$ $\quad$ $V_q = 14^3 \bmod 11 \equiv 3^3 \bmod 11 = 5$

$X_p = 11 \times (11^{-1} \bmod 3)$ $\quad$ $X_q = 3 \times (3^{-1} \bmod 11)$

$X_p = 11 \times (2^{-1} \bmod 3) = 11 \times 2 = 22$ $\quad$ $X_q = 3 \times (3^{-1} \bmod 11) = 3 \times 4 = 12$

$M = (V_pX_p + V_qX_q) \bmod n = (2.22 + 5.12) \bmod 33 = (44+60) \bmod 33 = 5$

$M = 5$

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ASYMMETRIC KEY CIPHERS

Security of RSA

❖ Brute force

This involves trying all possible private keys.

❖ Mathematical attacks

There are several approaches, all equivalent in effort to factoring the product of two primes.

❖ Timing attacks

- These depend on the running time of the decryption algorithm.
- Countermeasures to prevent timing attacks are (i) Constant exponentiation time, (ii) Random delay, and (iii) Blinding

❖ Chosen ciphertext attacks

❖ This type of attack exploits properties of the RSA algorithm.

Factoring Algorithms:

❖ Pollard's p-1 algorithm

❖ Pollard's rho algorithm

❖ Elliptic Curve Factorization Method (ECM)

❖ Quadratic Sieve: It is fastest method for factoring integers less than 110 digits.

❖ Generalized number field sieve: It is the most efficient method for factoring integers larger than 110 digits.

Generalized number field sieve and Quadratic sieve are the fastest algorithms and hence mostly used for performing mathematical attacks on RSA.

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Key Distribution

Keys distribution can be categorized into public key distribution and secret key distribution.

To distribute public keys there are various schemes like

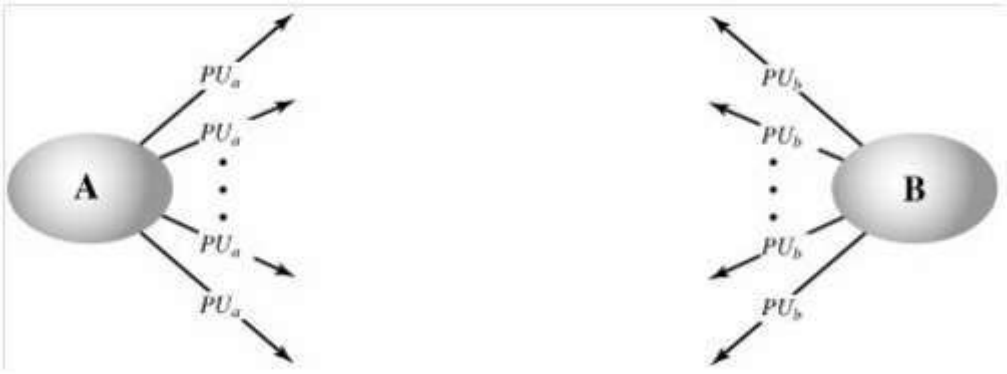
1.Public announcement

2.Publicly available directory

3.Public key authority

4. Public key certificates

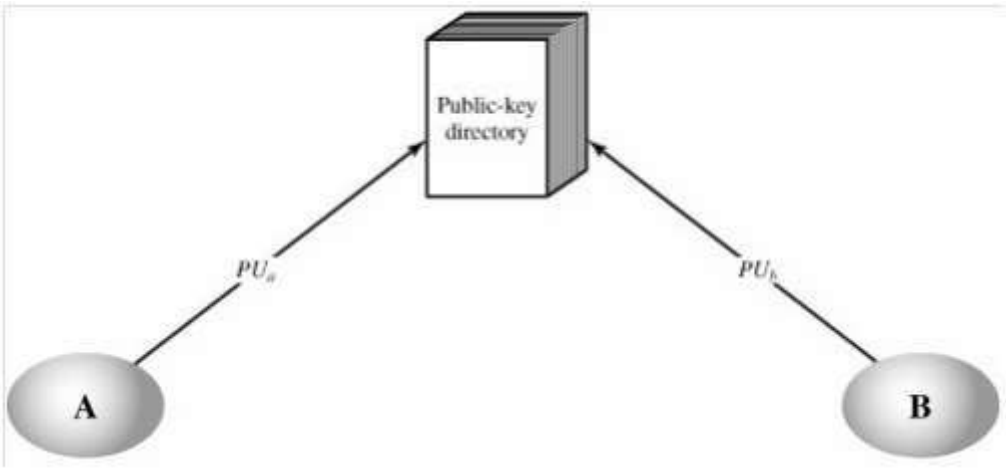
Public Announcement of Public Keys



In this distribution scheme the public keys are broadcasted to the large community. The limitation in this public announcement is that anyone can announce the key. Similarly a person A can pretend to be another person B and can publicly announce a public key for that user B. So when all users are sending the messages encrypted using the public key of user B it is actually can be decrypted only by user A.

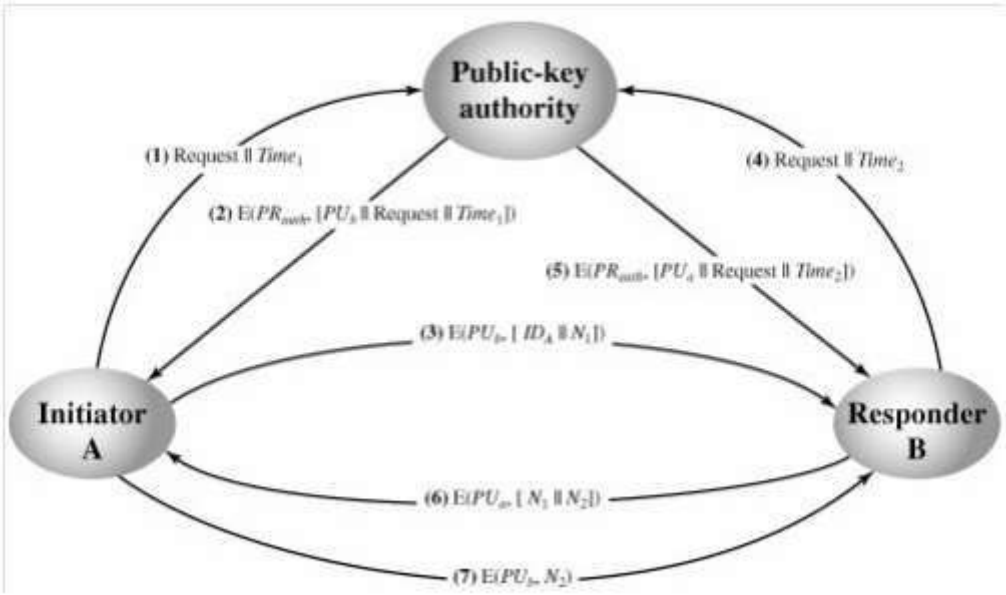
Publicly Available Directory

To overcome the drawback of the public announcement all users can share their public keys to the public directory which is maintained by an authority. Initially all users has to register in this public directory and each user is authenticated by the authority to access the public directory. So, it is not possible to perform masquerade attack. i.e., user A cannot send another person B's key. At any point of time the keys can be replaced by the concerned user. This scheme also has a drawback that an adversary can compromise the private key shared between the user and public authority.



Public-Key Authority

To avoid the weakness in the public directory scheme, each participant has to request the public authority for user B's public key by sending a time stamped request message. After receiving the request message, public key authority sends a response as encrypted message to the user A. Encryption is done using authority's private key which authenticates that the message is sent by the authority. Encrypted message has B's public key, and the user A's request message. Now if A wants to communicate with user B , user A will send an encrypted message to user B with a nonce and identifier of user A.

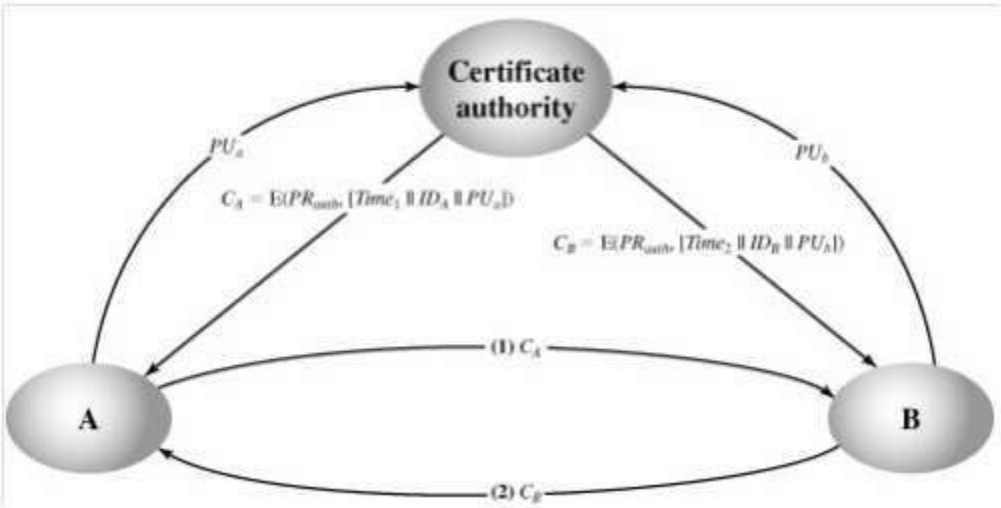


After receiving user B will send the reply as an encrypted message with the nonce sent by A and a new nonce value to uniquely identify the communication and to make an assurance that user B has received the message correctly. Later A will also send the new nonce value encrypted to make an assurance that both intended users are only communicating.

Public-Key Certificates

In the public key authority, as each user has to contact authority in-order to get the public keys. So bottleneck problem occurs. To avoid this issue each user will share the public key to the central authority who is responsible for issuing certificates to the users. Each user will be provided with certificate. So if any user wants to communicate they will exchange the certificates to authenticate themselves. Certificate will have users public key and the central authority's private key.

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Each certificate has

CA = E(PRauth, [T||IDA||PUa])

where PRauth is the private key used by the authority and T is a timestamp.

This certificate can be verified by performing

D(PUauth, CA) = D(PUauth, E(PRauth, [T||IDA||PUa])) = (T||IDA||PUa)

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DIFFIE HELLMAN KEY EXCHANGE

Whittfield Diffie and Martin Hellman are called the inventors of public key cryptography.

Diffie Hellman key exchange is the first public key algorithm published in 1976. Diffie Hellman is a public key algorithm. It only used for key exchange. Does not used for encryption and decryption. It is based on discrete logarithm. It is widely used in security protocols and commercial products.

Primitive roots

Let 'P' be a prime then 'b' is a primitive root for P if the powers of b, 1, b, b2 , b3…. by including all of the residue classes mod P except 0. Hence there must be P – 1 power of b.

Example 1:

If P = 7, then 3 is a primitive root for P because the power of 3 are 1, 3, 2, 6, 4, 5.

1, 3, 32, 33, 34 , …

1, 3, 9 mod 7, 27 mod 7, 31 mod 7

1, 3, 2, 6, 4, 5.

Example 2:

If P = 13, then 2 is a primitive root for P because the power of 2 are

1, 2, 22, 23, 24 , 25 , 26 , …

1, 2, 4 mod 13, 8 mod 13, 16 mod 13, 32 mod 13m 128 mod 13, 256 mod 13

1, 2, 4, 8, 3, 6, 12, 11, 9, 5, 10, 7

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Diffie Hellman Algorithm

Let 'q ' be a prime.

Given α where $\alpha < q$ and α ıs a primitive root of q.

User 'A' key generation

Select prime key XA where XA < q.

Calculate public key YA where $YA = \alpha^{XA} \bmod q$.

User 'B' generation

Select private key XB where XB < q.

Calculate public key YB where $YB = \alpha^{XB} \bmod q$.

Generation of secret key by user 'A'

$K = (YB)^{XA} \bmod q$

Generation of secret key by user 'B'

$K = (YA)^{XB} \bmod q$

Derivation

$K = (YB)^{XA} \bmod q$

$= (\alpha^{XB} \bmod q)\ XA \bmod q$

$= (\alpha^{XB})^{XA} \bmod q$

$= (\alpha^{XA})^{XB} \bmod q$

$= (\alpha^{XA} \bmod q)^{XB} \bmod q$

$K = (YA)^{XB} \bmod q$

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Elliptic Curve Cryptography

A major issue with the use of Public-Key Cryptography, is the size of numbers used, and hence keys being stored. Recently, an alternate approach has emerged, elliptic curve cryptography (ECC), which performs the computations using elliptic curve arithmetic instead of integer or polynomial arithmetic.

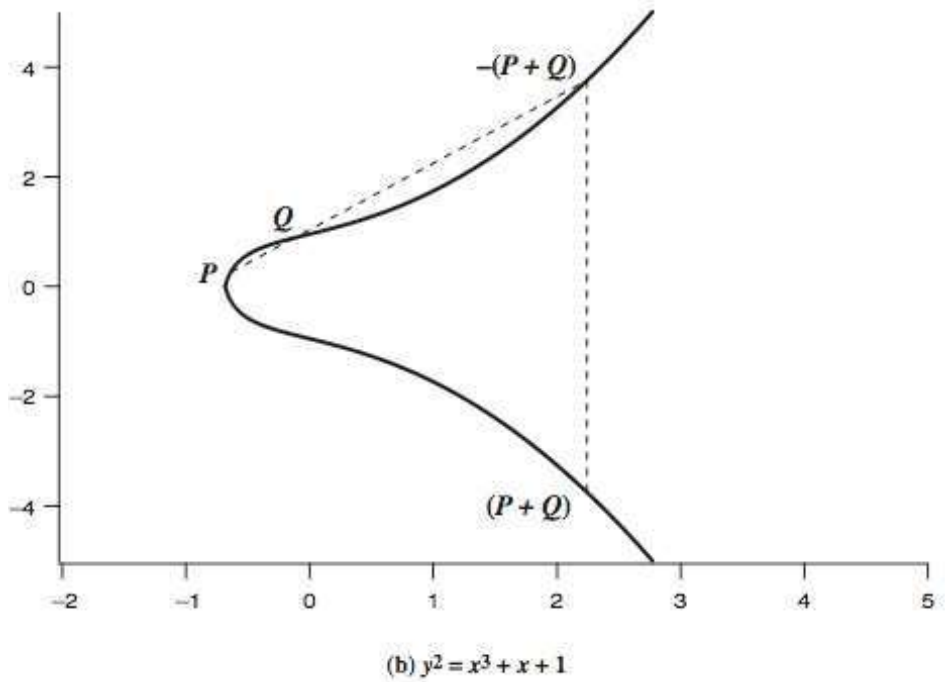
The principal attraction of ECC, compared to RSA, is that it appears to offer equal security for a far smaller key size, thereby reducing processing overhead.

Elliptic Curve

An elliptic curve is defined by an equation in two variables with coefficients. In general, cubic equations for elliptic curves take the following form, known as a Weierstrass equation:

$$y^2 + axy + by = x^3 + cx^2 + dx + e$$

where a, b, c, d, e are real numbers and x and y take on values in the real numbers.



(b) $y^2 = x^3 + x + 1$

Figure Example of Elliptic Curves

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For our purpose, it is sufficient to limit ourselves to equations of the form

$$y^2 = x^3 + ax + b$$

- where x,y,a,b are all real numbers, also define zero point O

➢ consider set of points E(a,b) that satisfy

➢ have addition operation for elliptic curve

- geometrically sum of P+Q is reflection of the intersection R

If three points on an elliptic curve lie on a straight line, their sum is O. hence define addition as:

1. O serves as the additive identity. Thus O = –O; for any point P on the elliptic curve, P + O = P. In what follows, we assume P <> O and Q <> O.
2. The negative of a point P is the point with the same x coordinate but the negative of the y coordinate; that is, if P = (x, y), then –P = (x, –y). These two points can be joined by a vertical line & that P + (–P) = P – P = O.

1. To add two points P and Q with different x coordinates, draw a straight line between them and find the third point of intersection R. There is a unique point R that is the point of intersection (unless the line is tangent to the curve at either P or Q, in which case we take R = P or R = Q, respectively). To form a group structure, we need to define addition on these three points as follows: P + Q = – R. ie. P + Q to be the mirror image (with respect to the x axis) of the third point of intersection as shown on slide.

2. The geometric interpretation of the preceding item also applies to two points, P and –P, with the same x coordinate. The points are joined by a vertical line, which can be viewed as also intersecting the curve at the infinity point. We therefore have P + (–P) = O, consistent with item (2).

3. To double a point Q, draw the tangent line and find the other point of intersection S. Then Q + Q = 2Q = –S.

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2. With the preceding list of rules, it can be shown that the set E(a, b) is an abelian group.

Elliptic curve arithmetic

Addition of Two Points

Let the points P = (x_1, y_1) and Q= (x_2, y_2) be in the elliptic group E_p(a, b), and O is the point at infinity. The rules for addition over the elliptic group E_p (a, b) are:

P+O=O+P=P

If x2= x1 and y2 = -y1, that is P = (x1, y1) and Q= (x2, y2)= (x_1, $-y_1$) = –P, then P+Q=O.

If Q = –P, then the sum P+Q = (x_3, y_3) is

given by: $x_3 = \lambda^2 - x_1 - x_2 \bmod p$

$y_3 = \lambda (x_1 - x_3) - y_1 \bmod p$

Where

$$\lambda \triangleq \begin{cases} \dfrac{y_2 - y_1}{x_2 - x_1}, & \text{if } P \neq Q \\ \dfrac{3x_1^2 + a}{2y_1}, & \text{if } P = Q \end{cases}$$

Subtraction of Two Points

Let the points P = (x1, y1) and Q = (x2, y2) be in the elliptic group Ep (a, b). The rules for subtraction over the elliptic group Ep (a, b) are:

P−Q=P+(−Q)=P+(inverse of Q), so the inverse of Q is Q(x2, y2)= Q(x2, -y2)

= Q(x2, p-y2).

Multiplication Points by a Constant

Let the points P = (x1, y1) and the integer be k in the elliptic group Ep (a, b). Then Pk= Add P with k times. e.g., 2P = P+P = (x1, y1) + (x1, y1) = (x3, y3).

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Point generation on Elliptic Curve

Let P = (3,10) ∈E23(1,1). Then 2P = (x3, y3) is equal to: 2P = P + P = (x1, y1) + (x1, y1) Since P=Q and x2 = x1, the values of λ, x3 and y3 are given by:

$$\lambda = \frac{3x_1^2 + a}{2y_1} \bmod p = \frac{3 \times 3^2 + 1}{2 \times 10} \bmod 23$$

$$= \frac{5}{20} \bmod 23 = 4^{-1} \bmod 23$$

Now using extended Euclidean formula: Here, quotient = q, remainder = r and $t = t_1 - qt_2$

Q	r_1	r_2	R	t_1	t_2	t
5	23	4	3	0	1	**-5**
1	4	3	1	1	-5	**6**
3	3	1	0	-5	6	**-23**
	1	**0**		**6**	**-23**	

Since, $r_1 = 1$. So that, $M_{1\text{-}1} = t_1$. The extended Euclidean Algorithm gives $t_1 = 6$. So, the multiplicative inverse (M^{-1}) is 6 mod 23 =6.

$x_3 = {}^2 - x_1 - x_2 = 6^2 - 3 - 3 \bmod 23 = 30 \bmod 23 = 7$

$y_3 = (x_1 - x_3) - y_1 = 6 \times (3-7) - 10 \bmod 23 = -34 \bmod 23 = 12$

Therefore $2P = (x_3, y_3) = (7, 12)$.

The multiplication kP is obtained by doing the elliptic curve addition operation k times by following the same additive rules.

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ECC Diffie-Hellman

The elliptic curve analog of Diffie-Hellman key exchange, which is a close analogy given elliptic curve multiplication equates to modulo exponentiation. Key exchange using elliptic curves can be done in the following manner.

First pick a large integer q, which is either a prime number p or an integer of the form 2^m and elliptic curve parameters a and b for Equation

$$y^2 \bmod p = (x^3 + ax + b) \bmod p$$

This defines the elliptic group of points $E_q(a, b)$. Next, pick a base point $G = (x_1, y_1)$ in $E_q(a, b)$ whose order is a very large value n. The order n of a point G on an elliptic curve is the smallest positive integer n such that $nG = O$. So $E_q(a, b)$ and G are parameters of the cryptosystem known to all participants. A key exchange between users A and B can then be accomplished as shown. To break this scheme, an attacker would need to be able to compute k given G and kG, which is assumed hard.

A key exchange between users A and B can be accomplished as follows

1. A selects an integer nA less than n. This is A's private key. A then generates a public key $P_A = n_A * G$; the public key is a point in Eq(a, b).
2. B similarly selects a private key n_B and computes a public key P_B.
3. A generates the secret key $k = n_A * P_B$. B generates the secret key

   $k = n_B * P_A$.

The two calculations in step 3 produce the same result because

$$n_A * P_B = n_A * (n_B * G) = n_B * (n_A * G) = n_B * P_A$$

To break this scheme, an attacker would need to be able to compute k given G and kG, which is assumed to be hard.As an example,6 take p = 211; Ep(0,-4), which is equivalent to the curve y2 = x3 - 4; and G = (2, 2). One can calculate that 240G = O.

- A's private key is $n_A = 121$, so A's public key is $P_A = 121(2, 2) = (115, 48)$.
- B's private key is $n_B = 203$, so B's public key is $203(2, 3) = (130, 203)$.
- The shared secret key is $121(130, 203) = 203(115, 48) = (161, 69)$.

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ECC Encryption/Decryption

Several approaches to encryption/decryption using elliptic curves have been analyzed in the literature. This one is an analog of the ElGamal public-key encryption algorithm. The sender must first encode any message M as a point on the elliptic curve P_m. Note that the ciphertext is a pair of points on the elliptic curve. The sender masks the message using random k, but also sends along a "clue" allowing the receiver who know the private-key to recover k and hence the message. For an attacker to recover the message, the attacker would have to compute k given G and kG, which is assumed hard.

Key Generation:

- Each user chooses private key $n_A<n$
- compute public key $P_A=n_AG$

Encryption:

- P_m : $C_m=\{kG, P_m+kP_b\}$, k random

Decryption:

C_m compute:

- $P_m+kP_b–n_B(kG) = P_m+k(n_BG)–n_B(kG) = P_m$

Let us consider a simple example. The global public elements are q = 257;

Eq(a, b) = $E_{257}(0, -4)$, which is equivalent to the curve $y^2 = x^3 - 4$; and G =(2, 2).

- Bob's private key is n_B = 101, and his public key is

$$P_B = n_BG = 101(2, 2)$$

$$= (197, 167).$$

- Alice wishes to send a message to Bob that is encoded in the elliptic point

$P_m = (112, 26)$.

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- Alice chooses random integer k = 41 and computes kG =41(2, 2) = (136, 128),

  $kP_B = 41(197, 167) = (68, 84)$

  $P_m + kP_B = (112, 26) + (68, 84) = (246, 174).$

- Alice sends the ciphertext

  $C_m = (C_1, C_2) = \{(136, 128),(246, 174)\}$ to Bob.

- Bob receives the ciphertext and computes

  $C_2 - n_B C_1 = (246, 174) - 101(136, 128)$

  $= (246, 174) - (68, 84)$

  $= (112, 26).$

ECC Security

The security of ECC depends on how difficult it is to determine k given kP and P. This is referred to as the elliptic curve logarithm problem. The fastest known technique for taking the elliptic curve logarithm is known as the Pollard rho method. Compared to factoring integers or polynomials, can use much smaller numbers for equivalent levels of security. Comparable Key Sizes for Equivalent Security shown in below table:

Symmetric scheme (key size in bits)	ECC-based scheme (size of n in bits)	RSA/DSA (modulus size in bits)
56	112	512
80	160	1024
112	224	2048
128	256	3072
192	384	7680
256	512	15360

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ASSIGNMENT UNIT III

S.No	Questions	CO	K
SET 1			
1.	A number n=561 passes **Fermat's Little Theorem** for base a=2 Prove that 561 is composite. Explain why it is a **Carmichael number**. Show how **Miller–Rabin primality test** detects it. .	CO3	K5
2.	Design a hybrid primality test: ▪ Combine Fermat + Miller-Rabin ▪ Analyze worst-case time complexity ▪ Evaluate probability of false positives after 5 iterations	CO3	K6
SET 2			
3.	Solve and justify uniqueness: $x \equiv 2 \pmod 3, \quad x \equiv 3 \pmod 5, \quad x \equiv 2 \pmod 7$ Verify solution using Chinese Reminder Theorem.	CO3	K4
4.	Show how CRT reduces decryption complexity from $O(n^3)$to smaller components. Compute decryption using CRT for p=11, q=13, d=7, C=9.	CO3	K4

Real life case study

Whether symmetric algorithm or Public key cryptography is widely used in current scenario. Explain

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ASSIGNMENT UNIT III

S.No	Questions	CO	K
SET 3			
5.	Given p = 23, g =5. Alice a = 6, Bob b = 15. ▪ Compute Shared key using Diffie-Hellman Key Exchange. ▪ Simulate a Man-in-the-Middle Attack. ▪ Show mathematically how attacker derives different keys.	CO3	K3, K4
6.	Given p = 23, g = 5, x = 6 (private key) Message M = 10, random k = 7. ▪ Compute public key. ▪ Encrypt using El-Gamal Encryption. ▪ Decrypt and verify correctness.	CO3	K5
SET 4			
7.	Post-Quantum Analysis - Evaluate impact of **Shor's Algorithm** on: RSA Diffie-Hellman ECC	CO3	K5
8.	Compare Hardness: Integer factorization vs elliptic curve discrete log problem Why **Elliptic Curve Cryptography** achieves same security with smaller keys?	CO3	K5
SET 5			
9.	Construct an algorithm for efficient modular exponentiation and analyze its time complexity.	CO3	K6
10.	Propose a secure key distribution framework using **Public Key Infrastructure** and blockchain.	CO3	K6

AI Based Examples

1. AI-Assisted Primality Testing and Factorization

Concept:
Primes, Primality Testing, Factorization

AI Example:
An AI-assisted primality testing and factorization.

Scenario:
Students use an AI tool and asked to explain whether they are prime and demonstrate suitable primality-testing/factorization steps.

AI analyzes:
Verify the AI-generated solution manually.

Real-life applications:
Educational purposes to understand the primality testing.

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2. AI Tutor for Euler's Totient, Fermat's and Euler's Theorems

Concept: Euler's Totient Function, Fermat's & Euler's Theorem

Usage:
AI generates numerical problems of different difficulty levels and provides step-by-step hints rather than directly giving answers. Students solve first and compare their solutions with AI explanations.

3. AI-Based RSA Key Generation Simulator

Concept: RSA Cryptosystem, Key Distribution

Usage:
Students use AI to generate a small RSA example using selected primes, calculate n, $\phi(n)$, public/private keys and demonstrate encryption/decryption. AI is also asked to explain each step.

4. AI Simulation of Diffie–Hellman Key Exchange

Concept: Diffie–Hellman Key Exchange

Usage:
AI acts as Alice and Bob and generates a guided numerical simulation. Students independently calculate the public values and shared secret and identify whether both parties obtain the same key.

5. AI-Based RSA vs ECC Security Analyst

Concept: Elliptic Curve Arithmetic, ECC, RSA

Usage:
Students ask AI to compare RSA and ECC for a given application such as mobile banking, IoT or secure messaging. They evaluate key size, computational requirements, security assumptions and suitability.

TWO MARKS Q & A

1. Define one-way function. (CO3, K1)

A one-way function is one that maps a domain into a range such that every function value has a unique inverse, with the condition that the calculation of the function is easy whereas the calculation of the inverse is infeasible:

$Y = f(X)$ easy

$X = f^{-1}(Y)$ infeasible

2. Define trap-door one-way function. (CO3, K1)

It is one that is easy to calculate in one direction and infeasible to calculate in the other direction unless certain additional information is known. With the additional information the inverse can be calculated in polynomial time.

$Y = f_k(X)$ easy, if k and X are known

$X = f^{-1}_k(Y)$ easy, if k and Y are known

$X = f^{-1}_k(Y)$ infeasible, if Y is known but k is not known

3. Compare secret key and public key cryptography. (CO3, K2)

S.No.	Secret Key Cryptography	Public Key Cryptography
1	The key is shared between sender and receiver. It is not known to others.	Private key of b user is known only to that user. Public key of b user is known to group of users.
2	Encryption and decryption algorithms use same key. Hence it is known bs symmetric key encryption.	Encryption and decryption algorithms use different keys. Hence it is known as asymmetric key encryption.

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4. State Fermat's theorem. (CO3, K1)

If p is prime and a is a positive integer not divisible by p, then $a^{p-1} \equiv 1 \pmod{p}$.

Example: $7^{18} \equiv 1 \pmod{19}$

5. Define Euler's totient function. (CO3, K1)

It is defined as the number of positive integers less than n and relatively prime to n. It is denoted by φ (n). Example: φ (11) = 10, φ (10) = 4

6. State Euler's theorem. (CO3, K1)

If b and n are relatively prime, then $a^{\varphi(n)} \equiv 1 \pmod{n}$. Example: $13^8 \equiv 1 \pmod{20}$

7. Define primitive roots. (CO3, K1)

If a is a primitive root of n, then its powers a, a^2, a^3, …, $a^{\varphi(n)}$ are distinct (mod n) and are all relatively prime to n. Moreover $a^{\varphi(n)} \equiv 1 \pmod{n}$.

8. Define discrete logarithm. (CO3, K1)

$$b \equiv a^i \pmod{p} \text{ where } 0 \leq i \leq (p-1)$$

This exponent i is referred to as the discrete logarithm of the number b for the base a (mod p). This value is denoted as $\text{dlog}_{a,p}(b)$.

9. The Miller-Rabin test can determine if a number is not prime but cannot determine if a number is prime. How can such an algorithm be used to test for primality? (CO3, K2)

If the number of Miller-Rabin tests t=1, the probability that a nonprime number will pass the test is less than 0.25. If the number of tests increased to t=10, then probability that a nonprime number will pass all ten tests is less than 10^{-6}. In general getting false positive is $(1/4)^t$. So by choosing t different value of a, one can close to 100% confidently determine a number as prime.

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10. List out the three broad categories of applications of public-key cryptosystems (CO3, K2)

- ❖ Encryption and decryption
- ❖ Digital Signature
- ❖ Key Exchange

11. How is authentication and confidentiality achieved by asymmetric encryption system? (CO3, K2)

- ❖ Confidentiality is achieved when the sender encrypts the message with the recipient's public key.
- ❖ Authentication is achieved when the sender signs a message with its (sender's) private key.

12. What is the idea behind the security of RSA?

The security of RSA lies in the fact that Integer Factorization (factoring an integer into their prime factors) is hard.

13. List out the different primality-testing algorithms. (CO3, K2)

- ❖ Fermat primality test
- ❖ Miller-Rabin primality test
- ❖ Solovay-Strassen primality test

14. List out the different types of attacks on RSA? (CO3, K2)

- ❖ Low Exponent attack
- ❖ Short plaintext attack
- ❖ Timing attack
- ❖ Mathematical attack (Factoring)

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15. List out the different factorization algorithms.

- ❖ Quadratic Sieve
- ❖ Generalized number field sieve
- ❖ Elliptic Curve Factorization Method (ECM)

16. Define man-in-the middle-attack. (CO3, K1)

An attacker interposes during key exchange, acting as the client to the server and as the server to the client. This attack involves persuading the client and server to believe that they are talking to each other when in fact the communication is going through an intermediate attacker.

17. Define an elliptic curve. (CO3, K1)

An elliptic curve is defined by an equation in two variables with coefficients. For cryptography, the variables and coefficients are restricted to elements in a finite field.

$$y^2 = x^3 + ax + b$$

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18. Perform encryption and decryption using RSA Alg. for the following.

P=7; q=11; e=17;M=8.

n = pq

n = 7*11=77

Φ(n)=(p-1) (q-1)

=6*10 = 60

e =17 d =27

C = M^e mod n

C = 817 mod 77

= 57

M = Cd mod n

= 5727 mod 77 = 8

19. Specify the applications of the public key cryptosystem?

The applications of the public-key cryptosystem can classified as follows

Encryption/Decryption: The sender encrypts a message with the recipient's public key.

Digital signature: The sender "signs" a message with its private key. Signing is achieved by a cryptographic algorithm applied to a message or to a small block of data that is a function of the message.

Key Exchange: Two sides cooperate to exchange a session key. Several different approaches are possible, involving the private key(s) of one or both parties.

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20. List four general characteristics of schema for the distribution of the

public key?

The four general characteristics for the distribution of the public key are

Public announcement

Publicly available directory

Public-key authority

Public-key certificate

21. What is a public key certificate?

The public key certificate is that used by participants to exchange keys without contacting a public key authority, in a way that is as reliable as if the keys were obtained directly from the public-key authority. Each certificate contains a public key and other information, is created by a certificate authority, and is given to a participant with the matching private key.

22. What is key distribution center?

A key distribution center is responsible for distributing keys to pairs of users such as hosts, processes, applications. Each user must share a unique key with the key distribution center for purposes of key distribution.

23. List out the attacks to RSA.

Brute force - Trying all possible private keys.

Mathematical attacks - The approaches to factor the product of two prime numbers.

Timing attack - Depends on the running time of the decryption algorithm.

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PART B

1. Users Alice and Bob use the Diffie Hellman key exchange technique with common prime q = 83 and primitive root α = 5.

(i) If Alice has a private key as 6, what is public key of Alice?

(ii) If Bob has a private key as 10 , what is the public key of Bob?

(iii) What is the shared secret key?

2. Users A and B use the Diffie-Hellman key exchange technique, a common prime p=71 and a primitive root g=7 are used. If user A has private key X A =5, what is A's public key Y A ? If user B has private key X B =12, what is B's public key Y B ? What is the shared secret key?

3. Explain briefly about man-in-the-middle man attack.

4. Find the secret key shared between user A and user B using Diffie Hellman algorithm for the following q=353 , α = 3, XA = 45 and XB = 50

5. Explain ElGamal Cryptosystem

6. Explain key distribution techniques.

7. Using Elliptic curve encryption/decryption scheme, key exchange between users A and B is accomplished. The cryptosystem parameters are, elliptic group of points E 11 (1,6) and point G on the elliptic curve is G=(2,7). B's secret key is n B =7. Now when (i) A wishes to encrypt the message Pm=(10,9) and chooses the random value K=3. Determine the ciphertext Cm. Find out B's public key PB.

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8. State Chinese Remainder Theorem and find X for the given set of congruent equations using CRT

$X \equiv 1 \pmod 5$

$X \equiv 2 \pmod 7$

$X \equiv 3 \pmod 9$

$X \equiv 4 \pmod{11}$

9. Perform encryption and decryption using RSA algorithm. Given: p=5; q=11; e=3; M=9. Explain the strength of RSA.

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GATE QUESTIONS

1. RSA Cryptosystem – GATE CSE 2019

Question:
In an RSA cryptosystem, the value of the public modulus parameter n is 3007. If it is also known that (n)=2880, where ϕ denotes Euler's Totient Function, then the prime factor of n which is greater than 50 is _____.

Answer: 97

Explanation:
Given n=3007 and ϕ(n)=2880. Since n=p×q and ϕ(n)=(p−1)(q−1), we can factor n to find its prime factors. Testing for divisibility, we find 3007=31×97. Among these, the prime factor greater than 50 is 97.

2. RSA Cryptosystem – GATE CSE 2017

Question:
In an RSA cryptosystem, a participant A uses two prime numbers p=13 and q=17 to generate her public and private keys. If the public key of A is 35, then the private key of A is _____.
Answer: 11
Explanation:
Compute n=p×q=13×17=221 and ϕ(n)=(p−1)(q−1)=12×16=192. The private key d is the modular multiplicative inverse of e=35 modulo ϕ(n)=192. Solving 35×d≡1mod 192, we find d=11.

3. Diffie-Hellman Key Exchange – GATE CSE 2005

Question:
Suppose that two parties A and B wish to set up a common secret key (D-H key) between themselves using the Diffie-Hellman key exchange technique. They agree on 7 as the modulus and 3 as the primitive root. Party A chooses 2 and party B chooses 5 as their respective secrets. Their D-H key is:
Options:
A. 3
B. 4
C. 5
D. 6
Answer: B. 4
Explanation:
Public parameters: modulus p=, base g=3. Private keys: A chooses a=2, B chooses b=5. A computes A=ga mod p=32 mod 7=2, B computes B=gbmod p=35mod 7=5 . Shared secret: A computes s=Bamod p=52mod 7=4, B computes s=Abmod p=25mod 7=4. So, the shared key is 4.

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4. Euler's Totient Function – GATE CSE 2005

Question:

If ϕ is Euler's Totient function, then $\phi(92)=$ _____.

Answer: 44

Explanation:

The prime factorization of 92 is 22×23. Euler's Totient function $\phi(n)$ for $n=pa\times qb$ is $\phi(n)=n\times(1-(1/p))\times(1-(1/q))$. Thus,

$\phi(92)=92\times(1-(1/2))\times(1-(1/23))=92\times(1/2)\times(22/23)=44$.

5. Chinese Remainder Theorem – GATE CSE 2007

Question:

Find the smallest positive integer x such that:

$x\equiv3 \mod 5$

$x\equiv4 \mod 7$

Answer: 24

Explanation:

Using the Chinese Remainder Theorem, we can find a unique solution modulo $5\times7=35$. Solving the congruences, we find $x=24$ satisfies both conditions.

6.Primality Testing – GATE CSE 2013

Question:

Which of the following numbers is a prime?

Options:

A. 91

B. 97

C. 111

D. 143

Answer: B. 97

Explanation:

Among the options, 97 is a prime number. The others can be factored:

$91 = 7 \times 13$

$111 = 3 \times 37$

$143 = 11 \times 13$

7. Fermat's Theorem – GATE CSE 2019

The value of 351 mod 5 is_____.

Answer

Correct answer is 2

10. Primality Testing – GATE CSE 2013

Which of the following is a probabilistic primality test?

A. Trial Division

B. AKS Primality Test

C. Fermat Primality Test

D. Sieve of Eratosthenes

Answer: C. Fermat Primality Test

Explanation: The Fermat Primality Test is a probabilistic method that can identify composite numbers but may falsely identify some composites (Carmichael numbers) as primes.

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NPTEL

- Cyber Security and Privacy
- Ethical Hacking
- Computational number theory and cryptography

COURSERA

- Cryptography
- Applied Cryptography
- Number theory and cryptography Cryptography and Information theory
- Asymmetric cryptography and key management Symmetric Cryptography

UDEMY

- Introduction to Cryptography Cryptography with python
- Applied Cryptography with Python
- Complete Cryptography master class

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Real Time Applications in day to day life and industry

- Secure Communication
- Diffie Hellman Algorithm can be used in
  - Transport Layer Security (TLS) / Secure Sockets Layer (SSL)
  - Public Key Infrastructure (PKI)
  - Internet Key Exchange (IKE)
  - Internet Protocol Security (IPSec)

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COMPLEX ENGINEERING PROBLEMS

Problem 1: Secure Key Exchange for a Distributed Communication System

Design a secure key-establishment mechanism for a distributed communication system in which two users need to establish a common secret key over an insecure public network without directly transmitting the secret key.

Tasks:

•Analyze mathematical requirements.
•Implement Diffie–Hellman.
•Analyze MITM and key management.
•Evaluate security, complexity & scalability.

Problem 2: Design of a Secure RSA-Based Data Exchange System

A financial application requires secure exchange of sensitive information between a client and server. Design an RSA-based asymmetric cryptographic solution that provides secure key generation, encryption/decryption and appropriate key management.

Tasks:

•Prime testing, factorization & φ(n).
•Generate RSA keys and encrypt/decrypt.
•Apply Fermat, Euler & CRT.
•Design key-management strategy & evaluate.

Problem 3: ECC-Based Secure Communication for Resource-Constrained Devices

Design a secure communication mechanism for an IoT/mobile environment where devices have limited computational power, memory and communication bandwidth. Evaluate whether **Elliptic Curve Cryptography (ECC)** is more suitable than RSA.

Tasks:

•Elliptic-curve arithmetic.
•Demonstrate ECC key generation.
•Compare RSA and ECC.
•Recommend solution based on engineering constraints.

SDG ACTIVITIES FOR THE COURSE

Activity 1: Interactive Cryptography Learning Activity
SDG: SDG 4-Quality Education
Course Topics Covered: Primes, Primality Testing, Euler's Totient Function, Fermat's & Euler's Theorem, CRT
Activity: Students work in small groups to solve progressively difficult number-theory problems using peer learning and AI-assisted verification. Each group prepares a simple learning resource explaining one mathematical concept and its cryptographic application.
Deliverables
•Problem-solving worksheet
• Student-created concept map/infographic
• AI-verified solutions
• Short group presentation

Activity 2: Secure Communication System Design
SDG: SDG 9 – Industry, Innovation and Infrastructure
Course Topics Covered: RSA, Key Distribution, Key Management, Diffie–Hellman, ECC
Activity: Students design a secure communication model for an IoT/industrial application. They compare RSA, Diffie–Hellman and ECC and select an appropriate asymmetric cryptographic technique considering security, computational resources and scalability.
Deliverables
•System architecture diagram
• Cryptographic algorithm flowchart
• RSA/Diffie–Hellman/ECC comparison
• Design justification and mini-report

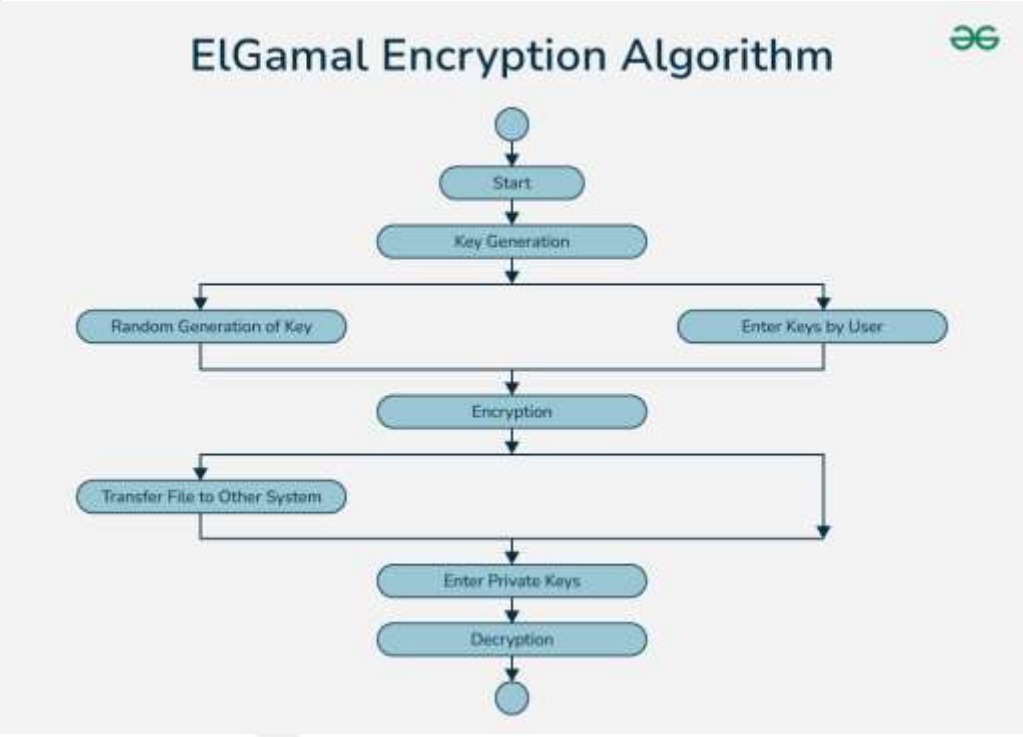
Activity 3: Secure Digital Services Case Study
SDG: SDG 16 – Peace, Justice and Strong Institutions
Course Topics Covered: RSA, Key Management, Diffie–Hellman, ECC, Cryptographic Security
Activity: Students analyze a scenario such as online banking, e-governance or secure digital communication and identify threats to confidentiality, authentication and key management. They propose an asymmetric cryptography-based security solution.
Deliverables
•Threat and risk analysis
• Secure communication architecture
• Key-management plan
• Security recommendations
• Group presentation

Contents beyond the Syllabus

ElGamal Encryption is a public-key cryptosystem. It uses asymmetric key encryption to communicate between two parties and encrypt the message. This cryptosystem is based on the difficulty of finding **discrete logarithms** in a cyclic group that is even if we know g^a and g^k, it is extremely difficult to compute g^{ak}.

Elgamal Cryptographic Algorithm

The ElGamal cryptographic algorithm is an asymmetric key encryption scheme based on the Diffie-Hellman key exchange. It was invented by Taher ElGamal in 1985. The algorithm is widely used for secure data transmission and has digital signatures and encryption applications.



Components of the ElGamal Algorithm

Key Generation:

Public Parameters: Select a large prime number p and a generator g of the multiplicative group Z^*_p.

Private Key: Select a private key x such that $1 \leq x \leq p-2$.

Public Key: Compute $h=g^x \bmod p$. The public key is (p,g,h) and the private key is x.

Encryption:

To encrypt a message M:

Choose a random integer k such that $1 \leq k \leq p-2$.

Compute $C_1 = g^k \bmod p$.

Compute $C_2 = M \cdot h^k \bmod p$.

The ciphertext is (c1,c2).

Decryption:

To decrypt the ciphertext (c1,c2) using the private key x:

Compute the shared secret $s = C_1^x \bmod p$.

Compute $s^{-1} \bmod p$ (the modular inverse of s).

Compute the original message $M = C_2 \cdot s^{-1} \bmod p$.

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ASSESSMENTSCHEDULE

Assessment	Proposed Date	Actual Date
IAT I	28.07.2026	04.08.2026
IAT II	31.08.2026	08.09.2026
MODEL	21.09.2026	28.09.2026

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Seminar Topics

1. **RSA Cryptosystem: From Mathematical Foundations to Secure Communication**
2. **Diffie–Hellman Key Exchange: Establishing a Secret over an Insecure Network**
3. **Elliptic Curve Cryptography: Why Smaller Keys Can Provide Strong Security**
4. **RSA vs ECC: Which Public-Key Algorithm is Better?**
5. **Chinese Remainder Theorem: Accelerating RSA Computation**
6. **Primality Testing and Prime Generation in Cryptography**
7. **Man-in-the-Middle Attack and Secure Key Exchange**
8. **Cryptographic Key Management in Modern Digital Systems**
9. **ECC Applications in IoT, Mobile Devices and Embedded Systems**
10. **Post-Quantum Cryptography: Threats to RSA and ECC**

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Mini Projects Suggestions

1. RSA Encryption and Decryption Implementation:

1. Develop a Secure Text Messenger using the RSA
2. Design a Credit Card Data Protection Application using the RSA securing the sensitive information of CVVs and card numbers.

2. Digital Signature Scheme Implementation:

1. Develop a Secure Document Signer using DSA (Digital Signature Algorithm).
2. Demonstrate a Hashed Password Cracker using the DSA.

3. Key Exchange Protocol Simulation:

1. Simulate a Secure Key Exchange Simulation in Client-Server Network Architecture.
2. Visualize the Step-by-Step Simulation of Secure Key Negotiation for Educational Use.

4. Secure Communication Application:

1. Develop a Secure Communication App for Android using Hybrid Cryptography (RSA + AES).
2. Simulate Self-Destructing Messages for Privacy-Focused Communication Apps.

5. Cryptographic Protocols Analysis:

1. Security Analysis of Diffie-Hellman Key Exchange under Adversarial Conditions.
2. Comparative Study of Symmetric vs. Asymmetric Cryptographic Protocols.

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5. Blockchain and Cryptocurrency:

1. Develop a Decentralized Voting System using Blockchain Technology.
2. Simulation of a Cryptocurrency System Inspired by Bitcoin.

6. Side-Channel Attacks Experimentation:

1. Timing Attack Vulnerabilities in RSA Cryptosystems: Experimental Evaluation.
2. Side-Channel Attack Simulation using Software-Based Leakage Models.

7.Homomorphic Encryption Demonstration:

1. Implementation and Demonstration of Homomorphic Encryption for Secure Computation.
2. Simulate and Perform Arithmetic Operations on Encrypted Data.

8. Post-Quantum Cryptography Exploration:

1. Explore Post-Quantum Cryptography for Future Secure Communication.
2. Implementation and Analysis of Quantum-Resistant Cryptographic Algorithms.

9. Security Analysis of Asymmetric Cryptography in IoT Devices:

1. Simulate and Analyze the Asymmetric Cryptographic Protocols in IoT Networks.
2. Develop a Threat Model and Security Assessment of IoT Systems using Asymmetric Encryption.

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TEXT BOOKS AND REFERENCES

TEXT BOOK:

1. William Stallings, Cryptography and Network Security: Principles and Practice, Pearson education 8th Edition, 2023.
2. Nina Godbole, Sunit Belapure, “Cyber Security: Understanding Cyber crimes, Computer Forensics and Legal Perspectives”, First Edition, Wiley India, 2011.

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2. Behrouz A. Ferouzan, Debdeep Mukhopadhyay, "Cryptography and Network Security", 3rd Edition, Tata Mc Graw Hill, 2015.
3. Wade Trappe, Lawrence C. Washington: Introduction to Cryptography with Coding Theory, 3rd Edition, 2020.
4. Charles Pfleeger, Shari Pfleeger, Jonathan Margulies, "Security in Computing", Fifth Edition, Prentice Hall, New Delhi, 2015.

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